📘 MODULE 20: DATA INPUTS

20.1 Concept

A **data input** tells Splunk *what to collect and how*. Inputs are defined in `inputs.conf` or through Settings → Add Data.

20.2 Full flow

```
DATA SOURCE → INPUT (inputs.conf) → FORWARDER → OUTPUTS.CONF → RECEIVER (port 9997)
            → INDEXER → INDEX
```

On a standalone lab, the input and the indexer are the same instance. In production, the UF holds `inputs.conf` and `outputs.conf`, and the indexer holds the receiving port and `indexes.conf`.

20.2b Input types at a glance

Input	Used for
Monitor	Files and directories (most common)
TCP / UDP	Network data, syslog
Scripted	Output of a script (commands, APIs)
Windows inputs	Event logs, registry, perfmon (via UF on Windows)
HEC	Applications sending JSON over HTTP (Module 24)
Modular / API inputs	Add-ons that pull from cloud/SaaS APIs

20.3 Monitor inputs

ini

```ini
[monitor:///var/log/apache2]
index = web
sourcetype = access_combined
whitelist = \.log$
blacklist = \.gz$
recursive = true
disabled = 0
```

Breakdown:

- `monitor://` + a **path** (file or directory). Three slashes: `://` + the leading `/`.
- `whitelist` / `blacklist`: **regex** on the file name to include or exclude.
- `index`, `sourcetype`: Module 4 and 5.
- Splunk remembers its position in each file using a **fishbucket**, so it won't re-read old lines. If you copy the same file under the same content, Splunk may skip it.

Verify:

```bash
sudo /opt/splunk/bin/splunk list monitor
```

Lists the files and directories being monitored.

```spl
index=_internal source=*splunkd.log* component=TailingProcessor
```

Shows file-reading messages and errors.
Common errors: permission denied on the file, wrong path, index doesn't exist, or the file wasn't updated, so there are no new events.

20.4 Network inputs: TCP, UDP, syslog

```ini
[udp://514]
index = network
sourcetype = syslog
connection_host = ip

[tcp://9514]
index = network
sourcetype = syslog
connection_host = ip
```

- `connection_host = ip` sets `host` to the **sender's IP** (other options: `dns`, `none`).
- Ports below 1024 need root privileges, so many labs use 5514 instead.
- **Best practice for production syslog:** a dedicated syslog server (rsyslog/syslog-ng) writes to files and a **UF monitors them**. This avoids losing UDP data when Splunk restarts.

20.5 Scripted inputs

```ini
[script://./bin/disk_usage.sh]
interval = 60
index = linux_metrics
sourcetype = disk_usage
disabled = 0
```

Splunk runs the script every 60 seconds and indexes its **stdout**. The script lives in the app's `bin/` folder, and it must be executable by the Splunk user. **Mistake:** a missing execute permission, or script output that doesn't have timestamps.

20.6 Windows inputs (Splunk perspective only)

```ini
[WinEventLog://Security]
index = windows
disabled = 0
start_from = oldest
whitelist = 4624,4625
```

- Runs on the **UF installed on Windows**.
- `WinEventLog://Security` is the event log channel.
- `whitelist` filters by event code, to limit volume (4624/4625 = logon success/failure).
- Other common channels are `Application`, `System`, and Sysmon (via `WinEventLog://Microsoft-Windows-Sysmon/Operational`).

20.7 HTTP Event Collector (preview)

Applications send JSON to `https://<host>:8088/services/collector/event` with a **token**. Full lab in Module 24.

20.8 APIs and modular inputs

Cloud and SaaS data (AWS, Microsoft 365, etc.) is usually pulled with a **vendor add-on** that runs on a Heavy Forwarder or a search head/IDM. You configure credentials and intervals in the add-on's UI. It's the same flow: input → index.

20.9 Enable the receiver (the other end)

Required for forwarders to send to your Splunk:

- **UI:** Settings → Forwarding and receiving → **Configure receiving** → **New Receiving Port** → **9997**.
- **CLI:**

```bash
sudo /opt/splunk/bin/splunk enable listen 9997 -auth admin:<password>
```

`enable listen 9997` opens the receiving port. `-auth` provides credentials.

- **Check:** `ss -tlnp | grep 9997` should show Splunk listening.

20.10 Hands-on lab: a monitored test log

1. Create a folder and a generator script:

```bash
```

```
mkdir -p /tmp/applogs
for i in $(seq 1 20); do echo "$(date '+%F %T') user=alice action=login status=success"
```

The loop writes 20 timestamped lines, one per second. Replace `alice` with other names on a second run for variety.
2. Make it readable by Splunk: `chmod -R a+rX /tmp/applogs`.
3. Add the input in `etc/system/local/inputs.conf`:

ini

```
[monitor:///tmp/applogs]
index = main
sourcetype = app_test
disabled = 0
```

4. Restart Splunk. Check with `splunk list monitor`.
5. Search:

spl

```
index=main sourcetype=app_test | stats count by user, status
```

6. Append another line and confirm it appears. If you delete and re-create the file, Splunk may treat it as already read (fishbucket).
7. On a lab only, to wipe data and re-test: stop Splunk and run `splunk clean eventdata -index main`. **This permanently deletes the index's data. Never do this in production.**

20.11 Troubleshooting inputs

Symptom	Check
No events	`splunk list monitor`, file permissions, correct path, time range (All time), correct index
Events in wrong index	`btool inputs list --debug` for conflicting stanzas
Duplicate or missing data	Fishbucket behavior, rotated files, same file path re-created
Timestamps wrong	`props.conf` time settings (Module 23)
UDP syslog gaps	Splunk restart or overload, so use file-based syslog collection

20.12 Interview Q&A

- **How does Splunk get data?** Monitor files, network ports, scripts, Windows inputs, HEC, and API/modular inputs.
- **What is the fishbucket?** Splunk's record of how far it has read each monitored file, so it doesn't re-index.

- **Why not receive UDP syslog directly in production?** Data loss when Splunk is down. A syslog server plus a UF is more reliable.
- **What does `whitelist` do in a monitor input?** Regex of file names to include.
- **What does `connection_host` do?** Sets how `host` is determined on network inputs.
- **Default receiving port?** 9997.

20.13 Revision

inputs.conf holds inputs · monitor/tcp/udp/script/WinEventLog/HEC · whitelist/blacklist are regex · fishbucket tracks file position · receiver port 9997 · verify with `list monitor` and `_internal`.